

LabWk4RegexEg

May 7, 2023

```
[1]: #URLs, e.g. http://www.uqlabs.co.uk
#Extend the above to accommodate variations, e.g. www.google.com, https://
    ↳uqlabs.co.uk, mailto:somebody@google.com

#Email addresses, e.g. fred@gmail.com,
#Extend the above to reject things like bob @ aol.com (spaces) or steve (no_
    ↳domain) or mary@aolcom (no period before .com)

#Dates in the form dd/mm/yyyy, e.g. 23/01/2021
#Extend the above to include dates in other forms, e.g. 23-Jan-2021, 2021/01/
    ↳23, 23-01-2021, etc.
```

```
[2]: import re
```

```
[3]: text3 = '''#Dates in the form dd/mm/yyyy, e.g. 23/01/2021
#Extend the above to include dates in other forms, e.g. 23-Jan-2021, 2021/01/
    ↳23, 23-01-2021, etc.
'''
```

```
[4]: pattern3 = '(0?[1-9] | [12] [0-9] | 3[01])'
```

```
[5]: print(re.sub(pattern3, "X", text3))
```

```
#Dates in the form dd/mm/yyyy, e.g. XX/X/XXX
#Extend the above to include dates in other forms, e.g. XX-Jan-XXX, XXX/X/XX,
XX-X-XXX, etc.
```

```
[6]: pattern3 = '(0?[1-9] | [12] [0-9] | 3[01])(\\|\\-|\\-)'
```

```
[7]: print(re.sub(pattern3, "X", text3))
```

```
#Dates in the form dd/mm/yyyy, e.g. XX2021
#Extend the above to include dates in other forms, e.g. XJan-2021, 20XX23,
XX2021, etc.
```

```
[8]: pattern3 = '(0?[1-9]|12)[0-9]|3[01])(\\|\\-)(0?[1-9]|1[012])'
```

```
[9]: print(re.sub(pattern3, "X", text3))
```

#Dates in the form dd/mm/yyyy, e.g. X/2021
#Extend the above to include dates in other forms, e.g. 23-Jan-2021, 20X/23,
X-2021, etc.

```
[10]: pattern3 = '(0?[1-9]|12)[0-9]|3[01])(\\|\\-)(0?[1-9]|1[012])(\\|\\-)\d{4}'
```

```
[11]: print(re.sub(pattern3, "X", text3))
```

#Dates in the form dd/mm/yyyy, e.g. X
#Extend the above to include dates in other forms, e.g. 23-Jan-2021, 2021/01/23,
X, etc.

```
[12]: text2 = '''somebody@google.com'''
```

```
[13]: pattern2 = '[a-zA-Z0-9._%+-]+@'
```

```
[14]: print(re.sub(pattern2, "X", text2))
```

Xgoogle.com

```
[15]: pattern2 = '[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.'
```

```
[16]: print(re.sub(pattern2, "X", text2))
```

Xcom

```
[17]: pattern2 = '[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
```

```
[18]: print(re.sub(pattern2, "X", text2))
```

X

```
[19]: text1 = '''URLs, e.g. http://www.uxlabs.co.uk
www.google.com
https://uxlabs.co.uk
mailto:somebody@google.com
'''
```

```
[20]: pattern1 = '((http|https)\\:\\\\|\\/?[a-zA-Z0-9\\.\\|\\/?\\: @\\- _#]+\\.
→([a-zA-Z]){2,6}([a-zA-Z0-9\\.\\&\\|\\/?\\: @\\- _#])*)'
```

```
[21]: print(re.sub(pattern1, "X", text1))
```

URLs, e.g. X

X

X

X

[]: